Question	Answer	Marks	Partial Marks
1	0.0001	**1**	Allow 1×10^{-4} or 1/10000
2	57.40 cao	**1**	
3	275	**1**	
4	38	**2**	**M1** for 180 – 61 – 81 or for angle [BAC =] 180 – 2 × 61 or better soi by 58 or for angle [BAD =] 180 – 61 – (180 – 81) or better soi by 20
5	1700	**1**	
6	$6\frac{2}{3}$ oe	**2**	**M1** for figs 666 or 667
7(a)	$\begin{pmatrix} 27 \\ -12 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} 27 \\ k \end{pmatrix}$ or for $\begin{pmatrix} k \\ -12 \end{pmatrix}$
7(b)	13	**2**	**M1** for $(-12)^2 + 5^2$ oe soi by 169
8	$9 + 4.5\pi$ oe	**3**	**M2** for $\frac{9\pi}{2}$ oe or **M1** for 9π oe
9(a)	–3	**1**	
9(b)	$22 - 5n$ oe final answer	**2**	**B1** for $k - 5n$ or for $22 - kn$ $k \neq 0$ as final answer or for $22 - 5n$ oe seen then further incorrect working
10	$6\frac{1}{6}$ cao	**3**	**B2** for $\frac{37}{6}$ or **M2** for $\frac{16}{6} + \frac{21}{6}$ or for $5\frac{4}{6} + \frac{3}{6}$ or **M1** for $\frac{8}{3} + \frac{7}{2}$ or for $2 + 3 + \frac{2}{3} + \frac{1}{2}$
11	16	**2**	**B1** for 4 seen
12(a)	1.42×10^{-11}	**2**	**B1** for figs 142 or 10^{-12} seen
12(b)	5.72×10^{7}	**2**	**M1** for figs 572

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Question	Answer	Marks	Partial Marks
13	20	**2**	**M1** for $\frac{360}{180-162}$ or for $\frac{180(n-2)}{n} = 162$
14	One of the following 4, 10, 10, 12, 14 5, 10, 10, 10, 15 6, 8, 10, 10, 16	**4**	**B3** for 3 of the 4 conditions met or **B2** for 2 met or **B1** for 1 met e.g. total = 50
15	Rotation 90° clockwise oe (4, 3)	**3**	**B1** for each
16(a)(i)	30	**1**	
16(a)(ii)	20	**2**	**B1** for [u.q. =] 37 or for [l.q. =] 17
16(a)(iii)	39	**2**	**B1** for 320 seen
16(a)(iv)	34 to 38	**2**	**B1** for 362 to 366 seen
16(b)	Correct histogram, correct widths, heights 8 cm, 12.4 cm, 1.9 cm	**3**	**B1** for each column If 0 scored **SC1** for frequency densities 8, 12.4, 1.9 seen
17(a)(i)	34	**1**	
17(a)(ii)	106	**1**	
17(a)(iii)	72	**1**	**FT** *their* **(a)(ii)** – *their* **(a)(i)** or 106 – *their* **(a)(i)**
17(b)	6.6	**2**	**M1** for $\frac{4.5}{9} = \frac{3.3}{XD}$ oe
18(a)(i)	9	**1**	
18(a)(ii)	21	**1**	**FT** from *their* **(a)(i)**
18(b)	$\frac{3-x}{2}$ oe final answer	**2**	**M1** for $y + 2x = 3$ or for $\frac{y}{2} = \frac{3}{2} - x$ or for $x = 3 - 2y$
18(c)	$-\frac{1}{2}$ oe	**3**	**B2** for $4x + 6 + 3 = 7$ or better or **M1** for $2(2x + 3) + 3$ [= 7]
18(d)	32	**2**	**M1** for [x =] h(5)

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Question	Answer	Marks	Partial Marks
19(a)	$11\sqrt{2}$ final answer	**2**	**B1** for $4\sqrt{2}$ or $7\sqrt{2}$
19(b)	$\sqrt{2}-1$ final answer	**2**	**M1** for $\times \dfrac{\sqrt{2}-1}{\sqrt{2}-1}$
20	$2\frac{2}{7}$ or $\frac{16}{7}$ oe	**3**	**M2** for $y = \dfrac{16}{\sqrt{x}}$ or for $y\sqrt{49} = 8\sqrt{4}$ oe or **M1** for $y = \dfrac{k}{\sqrt{x}}$
21(a)	$\dfrac{39}{x+1}$ or $\left(\dfrac{22}{x}+2\right)$	**1**	
21(b)	$\dfrac{39}{x+1} - \dfrac{22}{x} = 2$ oe OR $(x+1)\left(\dfrac{22}{x}+2\right) = 39$ oe	**M1**	
	$39x - 22(x+1) = 2x(x+1)$ oe OR $22 + 2x + \dfrac{22}{x} + 2 = 39$	**M1**	**FT** *their* **M1** if two fractions with *their* linear denominators May still be with common denominator **FT** *their* **M1** if product of 2 pairs of brackets = 39
	$2x^2 - 15x + 22 = 0$	**A1**	Brackets removed and terms collected with no errors or omissions
21(c)	$(2x-11)(x-2)$ [= 0]	**M2**	**M1** for $(2x+a)(x+b)$ [= 0] where $ab = 22$ or $a + 2b = -15$ or $x(2x-11) - 2(2x-11)$ [= 0] or $2x(x-2) - 11(x-2)$ [= 0]
	$\dfrac{11}{2}$ oe and 2	**A1**	
	[h =] 4 and [h =] 11	**A2**	**A1** for each

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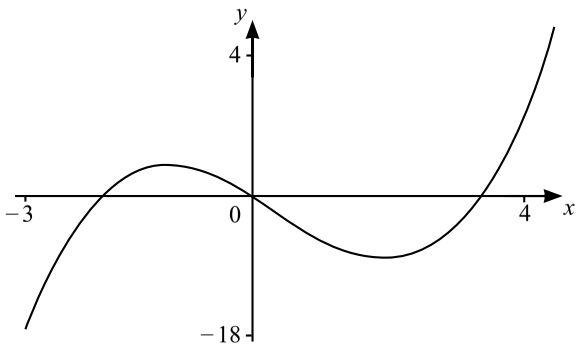
Question	Answer	Marks	Partial Marks
22	$\frac{16}{\sqrt{3}}$ or $\frac{16\sqrt{3}}{3}$	4	**B1** for $\cos 30 = \frac{\sqrt{3}}{2}$ **M2** for $\frac{8}{\cos 30}$ or **M1** for $\frac{8}{x} = \cos 30$
23	$\frac{25 - x}{(x - 4)(x + 3)}$ oe final answer	3	**M1** for denominator $(x – 4)(x + 3)$ or $x^2 – x – 12$ **B1** for $3(x + 3) – 4(x – 4)$ seen
24(a)	$(x – 2)^2 + 3$	2	**M1** for $(x – 2)^2$ seen
24(b)	(2, 3)	1	**FT** *their* **(a)**
25	$\frac{15}{2}$ oe	3	**M2** for $\frac{9}{x} = \frac{6}{5}$ oe or $x^2 = \frac{25}{36} \times 9^2$ oe or better or **M1** for $\sqrt{\frac{36}{25}}$ or $\sqrt{\frac{25}{36}}$ oe or better or for $\left(\frac{9}{x}\right)^2 = \frac{36}{25}$ or $\left(\frac{x}{9}\right)^2 = \frac{25}{36}$ oe or better
26(a)	Correct sketch and –2, 0 and 3 indicated on x-axis with graph going beyond intersections with x-axis  	3	**B2** for correct sketch without labels or **B1** for any positive cubic with two turning points
26(b)	$x^3 – x^2 – 6x$	3	**B2** for $x(x^2 + 2x – 3x – 6)$ or $(x^2 + 2x)(x – 3)$ or $(x + 2)(x^2 – 3x)$ or **B1** for $x^2 + 2x – 3x – 6$ or $x^2 + 2x$ or $x^2 – 3x$

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Question	Answer	Marks	Partial Marks
26(c)	(0, –1)	**5**	**B2** for $3x^2 - 2x - 6$ or **B1** for $3x^2$ or $2x$ or – 6 **B2** for $y = -5x - 1$ or **M1** for substituting 1 into *their* derivative and **M1dep** for $-6 = (\textit{their}\ -5) \times 1 + c$ or use of rise/run to find rise **FT** *their* **(b)** for the whole of **(c)**

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